

Zipf in Germany over time

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August 1, 2015

Abstract

In this paper, I use historical German population data to test Batty, et al. (2012)'s hypothesis about Zipf's law: that the Zipf coefficient corresponds to the economic integration level of the set of cities. Economically-integrated sets will have Zipf coefficients close to -1, and economically-disintegrated sets will have Zipf coefficients far from -1. This suggests that a perfectly integrated set will naturally grow toward the Zipfian distribution, and the reason Zipf's law may not hold in some contexts is not because it isn't the natural distribution; it could be because the set of cities chosen is not sufficiently integrated. Germany provides a natural experiment where the period of division (disintegration) following WWII can be compared with the period of reunification (integration) after the Berlin Wall is taken down. I find that the evidence broadly supports the economic integration theory, but there are some puzzling periods that aren't fully explained. The Zipf coefficient does generally seem to reflect the level of economic integration in Germany. Looking at Zipf's law this way emphasizes the importance of choosing cohesive sets when testing for Zipf's law and may provide an explanation for other studies that don't find a Zipfian distribution.

1 Theory

There is a phenomenon in urban economics called Zipf's law. It has held for decades across the world. It observes that people are distributed in cities across a country such that the biggest city has twice as many people as the second-biggest cities, three times as many people as the third-biggest cities, etc. It's very curious and elegant that this distribution exists, and no good explanation exists yet for why people would distribute themselves this way, instead of in any other way, especially over so many decades. The best explanation we have currently is the random growth theory, which notes that if cities all draw randomly from the same distribution of growth rates, then this will naturally over time give rise to Zipf's law.

The 2012 Nature article *There is More than a Power Law in Zipf* by Michael Batty, et al. suggests a new interpretation of Zipf's law. The theory is that the Zipf coefficient indicates the level of economic integration of a set of cities.

When we plot the rank and population of cities on doubly-logarithmic paper, and regress the $\log(\text{rank})$ on $\log(\text{population})$, a perfectly Zipfian distribution will have a slope coefficient of -1. Economically-integrated, or cohesive, sets should have Zipf coefficients close to -1, and economically-unintegrated sets should have Zipf coefficients that deviate from -1. In addition, the more economically-integrated a set is, the more positive the Zipf coefficient should be; the less economically-integrated a set is, the more negative the Zipf coefficient should be. While this doesn't explain the mechanism that creates Zipf's law, this would provide an interesting new way to think about Zipf's law and provide helpful insights.

2 Testing the theory

I test this theory by looking at the case of Germany and its Zipf coefficient over time, particularly how it changes during periods of division and unification. This is a good natural experiment to use because we know definitively the level of integration during the separation and unified periods (primarily the division following WWII and later reunification), so we can make a valid comparison. If the theory is true, then the Zipf coefficient for the set of German cities should deviate from -1 during the period of division, and then, it should return to -1 when Germany reunifies. Further, it should be close to -1 before World War II, and over time, overall, trend toward -1 because countries tend to develop and become more economically-integrated within themselves over time. For example, advancements in transportation technology and development of cities contribute to natural increases in countries's economic integration over time.

3 Historical background

During Germany's history, there have been two periods of unification, and lots of other historical events in between. I categorize the years into five periods.

Period 1 is from **1875-1885**. This is a period of integration following Germany's unification as a country in 1870.

Period 2 is from **1885-1919**. Germany is industrializing as the Second Industrial Revolution is from 1870-1914. It is also fighting WWI from 1914-1918. After WWI, in 1919, there is the Greater Berlin Act, which expands the administrative territory of Berlin. This period should be neither division nor division.

Period 3 is from **1919-1939**. This is another normal period. There is stability; the country is neither dividing nor integrating.

Period 4 is from **1939-1990**. This is the WWII and Cold War period. Germany participates in WWII from 1939-1945. After 1945, Germany is divided, and in 1961, the Berlin Wall is built to separate East and West Germany. In 1990, the Wall is taken down, and Germany reunifies. This is a period of division and disintegration.

Period 5 is from **1990-2012**. This is the modern period. This is a period of integration following Germany's reunification.

4 Data

To test this theory, I want data on the populations of the biggest cities in Germany for as many years as possible. Ideally, the administrative borders of the cities will consistently match the economic functional borders of the cities; however, this is not always the case. Sometimes the administrative unit will not keep up with the functional unit of the city, and this will affect the Zipf coefficient calculation.

I use data from several series of statistical publications. For the East German years, I use the Statistisches Jahrbuch der Deutschen Demokratischen Republik and the Statistical pocket book of the German Democratic Republic. For the West German years and unified Germany years, I use the Statistisches Jahrbuch für die Bundesrepublik Deutschland, Statistisches Jahrbuch für das Deutsche Reich, Statistisches Jahrbuch deutscher Städte, and Statistisches Jahrbuch deutscher Gemeinden. There is no continuous series of statistical publications, and I was not able to get the complete set for any one series.

4.1 Dataset

I make a few choices in building the dataset, which may affect the results.

1. I try to maintain consistency with the dates; however, sometimes the months differ. For example, years may have June and December data of the same year. I use the population at the end of a given year, instead of the beginning of the year. For example, Dec. 12, 1890 would be the 1890 data, instead of the 1891 data.

2. Some years are interpolated from census data; however, usually the data is updated from previous years.

3. I use population by *gemeinde*, which translates roughly to communities or municipalities. A *gemeinde* is the smallest administrative division and lowest level of official territorial division. I use *gemeinde* data, instead of *stadtkreise* (cities), *land* (state), or *kreis* (district), because it is the most specific level for which enough data is recorded and available.

4. I focus on the 10 biggest cities because Zipf's law usually holds only for the upper-tail of the distribution. I do also make a comparison with the top 15 and top 25 to check for robustness and confirm the same patterns hold.

5. For the separation years, I add the East and West Berlin populations together to get a Berlin (Total) number. I also add the East and West distributions together to get an artificial unified Germany distribution.

6. I use only modern Germany borders, so cities that used to be in the German empire (e.g. Königsberg now in Russia, Straßburg now in France) are excluded. This is because otherwise the changes in the Zipf coefficient may

be caused by some cities dropping out of the set. This only matters before 1955; after 1955, Germany has the modern borders we know today.

7. In 1929, Elberfeld, Barmen, and other neighboring villages are combined to form Wuppertal. For consistency, I add these two together for all the years.

8. Similarly, in 1929, Crefeld and Uerdingen combine to form Krefeld, so I add these together for all years.

9. From 1938-2007, Altona is combined with Hamburg. I do not combine Altona into Hamburg because I think Altona was not functionally part of the economic unit of Hamburg before 1938 and after 2007; between 1938-2007, Altona was part of Hamburg's functional unit, which is why it was administratively combined. It's possible that the administrative change lagged behind the economic changes. However, for comparison, I did calculate the coefficient with Altona combined with Hamburg for 1887 (Zipf coeff: -0.9959; compared to without, -0.9825) and 1936 (Zipf coeff: -1.0760; compared to without, -1.078003). It's difficult to tell if I'm artificially inflating Hamburg's population to bring it closer in line with ideal Zipf's or if the population at that time was actually distributed this way. I make a comparison graph and note that they follow the same pattern. Below is a graph of the Zipf coefficient of the top 10 cities in modern German borders with Altona combined with Hamburg, compared with when they're not combined, only for the years between 1875-1939 for which I have data.

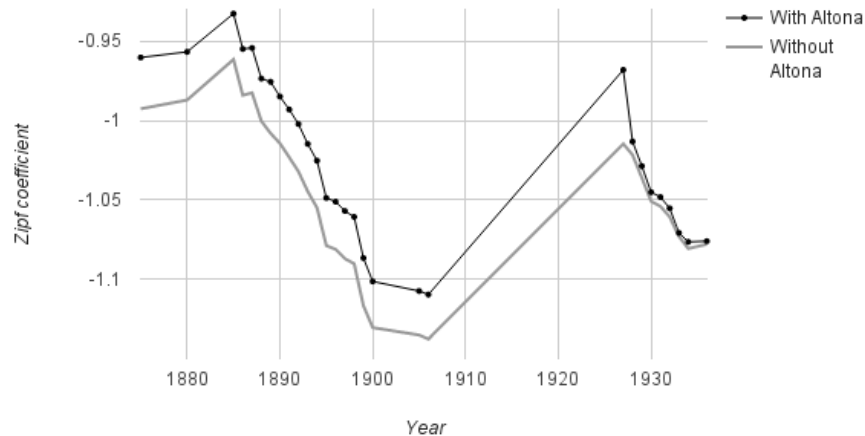


Figure 1: Comparison of Zipf coefficient calculated with and without Altona.

10. There is another city, Rixdorf, which is now part of Berlin; Rixdorf shows up within the top 25 cities in 1907 and 1910. For comparison, I calculate the coefficient with Rixdorf added to Berlin for 1910; combined, Zipf coeff: -1.1968,

and without, -1.2620. However, in the main dataset, I do not add Rixdorf to Berlin.

5 Findings

For each year, I rank the cities by population and graph the distribution double-logarithmically. I run the following linear regression and plot α_1 :

$$\ln(\text{rank}_c) = \alpha_0 + \alpha_1 \ln(\text{population}_c) + \varepsilon_c$$

A perfectly Zipfian distribution's regression line will have a slope of -1, i.e. ($\alpha_1 = -1$). I start with the year 1875 because that's when the first census of unified Germany is taken. I split the years into five main periods to analyze. Below is the graph of the Zipf coefficient of the top 10 cities in modern German borders over time with the five periods marked out.

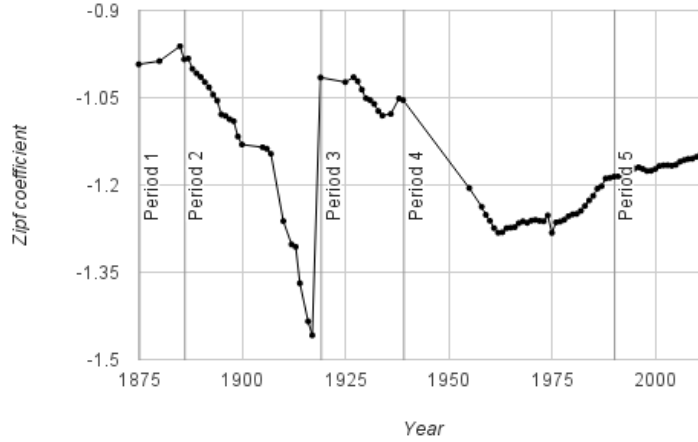


Figure 2: Zipf coefficient over the five periods.

5.0.1 Robustness

I calculate the Zipf coefficient for the top 15 and 25 biggest cities and compare these to the top 10 to check for robustness. The patterns are the same, with the top 10 being closer to -1 than the other two, as expected because Zipf's holds best for the upper-tail distribution. Below is the graph of the top 10, 15, and 25 biggest cities in modern German borders over time.

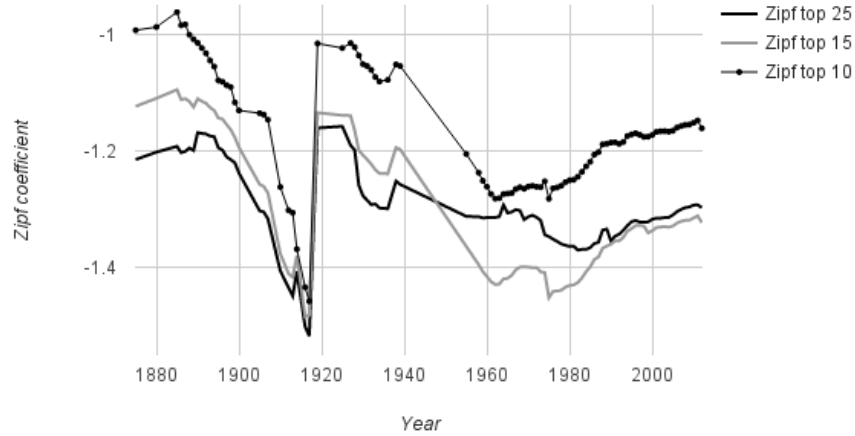


Figure 3: Comparison of the Zipf coefficient using top 25, 15, and 10 cities.

5.0.2 Modern German borders

I also compare the Zipf coefficient for the set of cities in modern German borders with the coefficient for the set of cities in the entire German empire when it existed before 1955. Prior to 1955, there are cities, such as Königsberg and Breslau, which are large top-10 cities in the German empire, but which are no longer part of Germany today (Königsberg is in Russia, Breslau is in Poland). After 1955, the borders were the same as those of modern Germany, so there is no difference. This supports the economic integration theory because the cities within modern Germany are more integrated than the entire area is. The cities no longer in Germany are on the sides of the German borders, and so it makes sense that they're less integrated with Germany. Below are the graph of the Zipf coefficient of the top 10 cities in the historical German empire, compared with the set of cities in modern German borders, and a map of the Germany's borders.

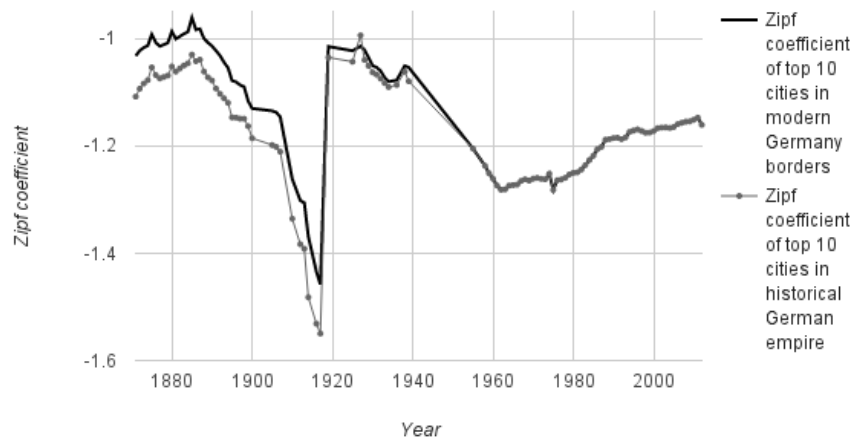


Figure 4: Comparison of the Zipf coefficient using modern and historical German borders.

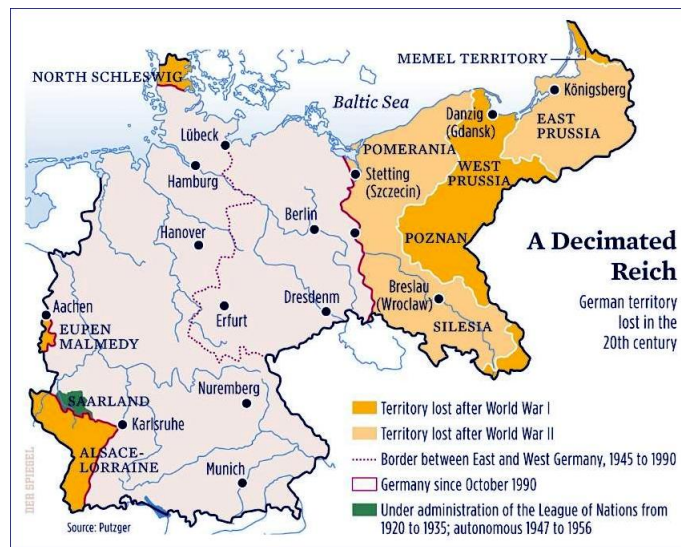


Figure 5: Map of German borders over time.

5.1 Period 1: 1875-1885

Germany unified as a country in 1870. During the period 1875-1885, the country is unifying and integrating. The dataset starts with a Zipf coefficient of -0.992614 in 1875. As predicted by the theory, the Zipf coefficient becomes more positive as the country integrates. It stays close to the ideal -1 throughout the period.

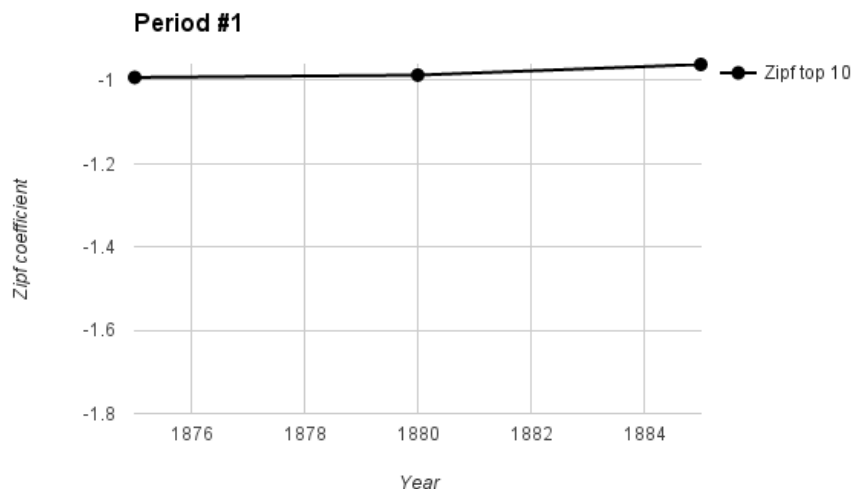


Figure 6: The Zipf coefficient during Period #1.

5.2 Period 2: 1886-1919

There is an inflection point at 1886; from 1886-1917, the Zipf coefficient decreases dramatically. Then, it increases back up in 1919. If the theory is correct, then this looks like a period of disintegration. However, it is unclear why Germany would be disintegrating during this period. Instead, I think other non-integration-related factors are causing this change. My best guess is a combination of two possible explanations: an administrative change and industrialization. During this period, the borders of Berlin are kept fixed, but the city grows all around it. In 1919, the Greater Berlin Act expands the borders of Berlin to allow the city to grow. When the administrative boundary is updated to reflect the functional boundary, the true distribution emerges.

I calculate the Zipf coefficient for the set of cities without Berlin. The Zipf coefficient for the set of cities without Berlin decreases during this period too, but the decrease is less pronounced, so the change in Berlin's borders doesn't account for all of the drop. After the Act, the Zipf coefficient jumps up close to -1 and makes a big difference, suggesting that it is a factor in the decrease

before. The graph below compares the Zipf coefficient during this period with and without Berlin.

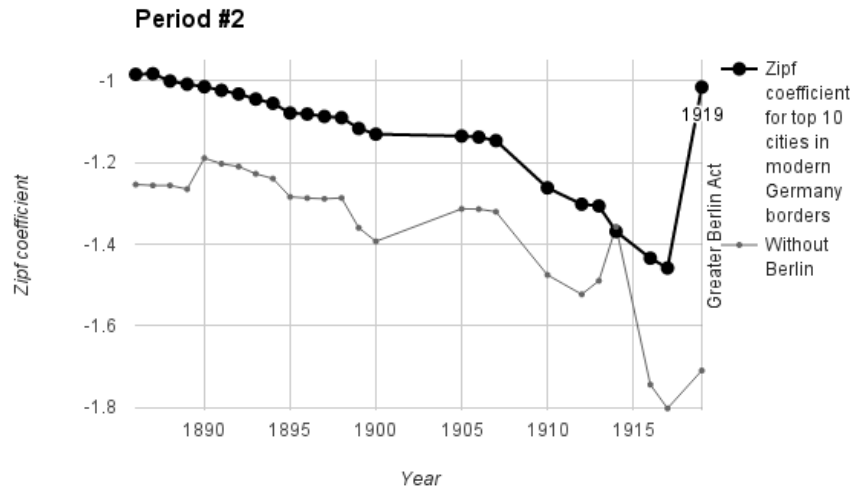


Figure 7: The Zipf coefficient during Period #2.

This period includes the Second Industrial Revolution, which was from 1870-1914, causing medium, industrial cities to grow faster than other cities and therefore change the population distribution. The medium cities are growing faster than Berlin. Essen is growing especially fast. Essen industrialized rapidly during this period; it's a coal-processing city located on the Ruhr. Similarly, Dusseldorf and Koln are on the Ruhr nearby and grow quickly this period. Below are a graph of the city growth rates within the top 10 cities over this period and a map of Germany.

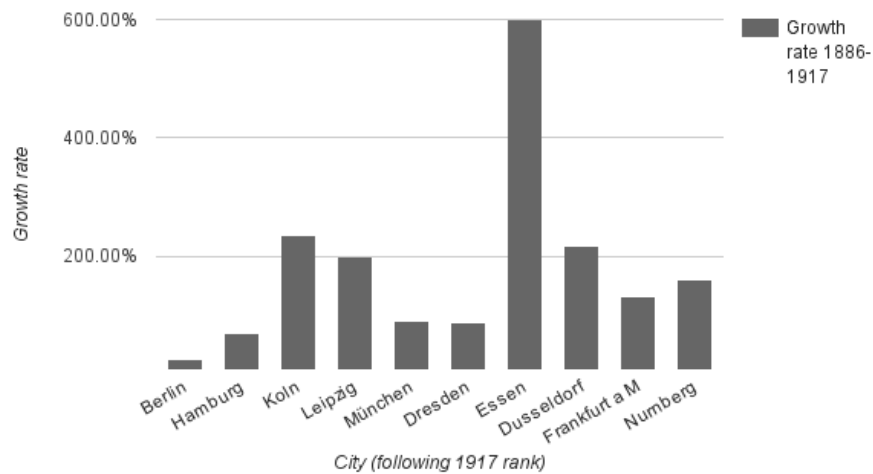


Figure 8: The city growth rates of the top 10 cities during this period.



Figure 9: Map of Germany.

5.3 Period 3: 1919-1939

During this period, the Zipf coefficient stays stable. It stays between -1 and -1.1. This supports the economic integration theory because there's no reason Germany would be disintegrating during this time. This is the normal or natural distribution. Below is the graph of the Zipf coefficient during this period.

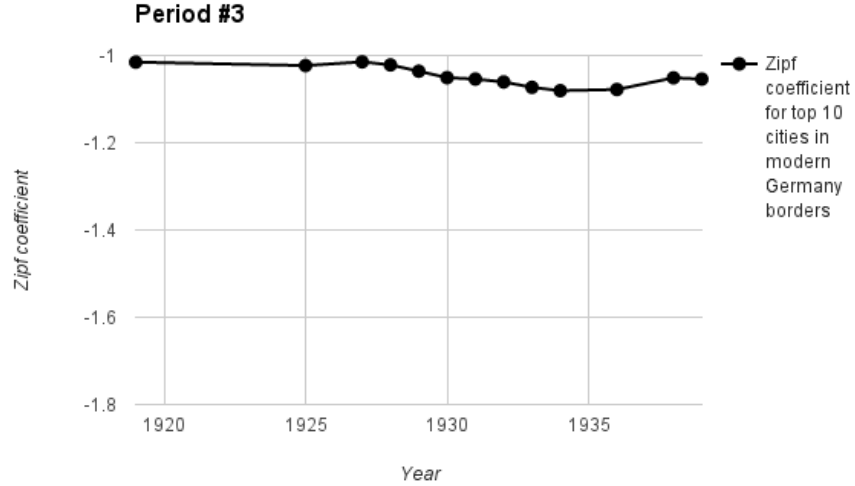


Figure 10: The Zipf coefficient during Period #3.

5.4 Period 4: 1939-1990

After the period of stability, sometime after 1939, the Zipf coefficient begins to decrease again. Unfortunately, I don't have many data points due to lack of data sources. World War II was from 1939-1945. After the war, Germany was divided into four occupation zones. The decreasing Zipf coefficient corresponding with this period of division supports the economic integration theory. The Zipf coefficient reaches its lowest right after the Berlin Wall is built. There is a small change in 1974-75, which corresponds to a territorial reorganization, including an expansion of Koln. Unexpectedly, the Zipf coefficient starts to grow closer to -1 at the end of this period before Germany is officially unified. Perhaps this is the result of increased immigration into West Germany during the 1980s when the Eastern bloc gradually opened.

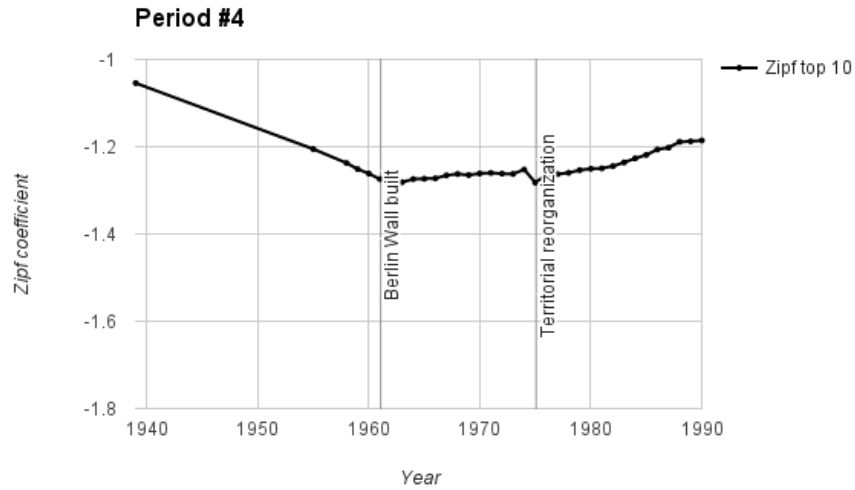


Figure 11: The Zipf coefficient during Period #4.

5.5 Period 5: 1991-2012

The Berlin Wall is taken down, and Germany is reunified, in 1990. This is a period of clear reintegration, and the Zipf coefficient corresponds by steadily moving upward toward -1. This is strong evidence in support of the economic integration theory.

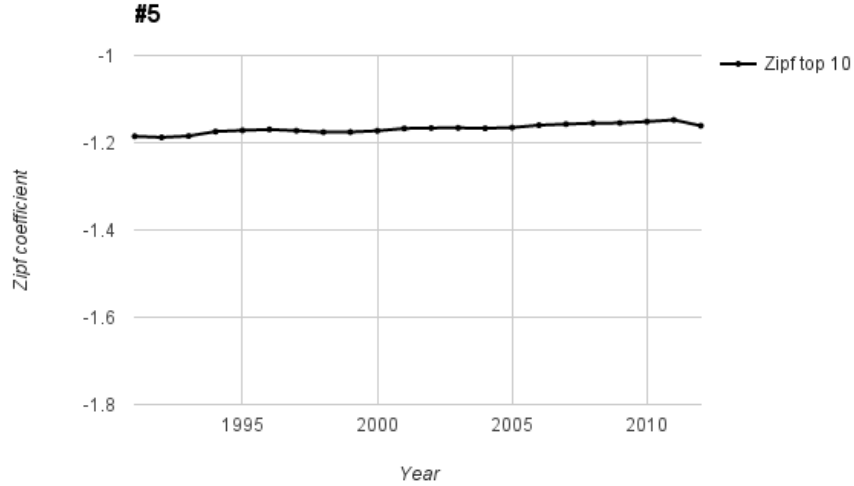


Figure 12: The Zipf coefficient during Period #5.

6 Conclusion

The evidence from Germany seems to broadly support the economic integration interpretation of Zipf's law. The Zipf coefficient generally reflects the level of integration in the country; it decreases during the period of division and increases during periods of unification. Acceptance of this interpretation predicts that as Germany continues to integrate and return to normal or its natural distribution, its Zipf coefficient will continue to get closer to -1. The strength of my evidence is weakened because my data is not comprehensive, and it's hard to say when the administrative units adequately reflect the actual economic units of the cities, and when they don't. There are also a couple puzzling periods. It would be nice to be able to graph the entire German population distribution and calculate a continuous Zipf's over time or re-calculate the Zipf coefficients according to the natural cities, following Jiang, et al. (2010, 2014). If this theory is proven correct, it may help explain Dittmar (2012)'s results, in that the Western European set he uses may have not been a cohesive set and over time, become integrated, which explains how Zipf's law emerges. This theory also predicts that the Zipf coefficient for Korea will follow a similar pattern when it unifies. While the economic integration theory does not explain why Zipf's law exists, it does provide insight into the Zipf coefficient as an indicator of economic integration, which could be related to the ultimate explanation.